

ORIGINAL RESEARCH

Serum neurofilament light chain predicts long-term prognosis in Guillain-Barré syndrome patients

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ABSTRACT

Objective To study baseline serum neurofilament light chain (sNfL) levels as a prognostic biomarker in Guillain-Barré syndrome (GBS).

Methods We measured NfL in serum (98 samples) and cerebrospinal fluid (CSF) (24 samples) of patients with GBS prospectively included in the International GBS Outcome Study (IGOS) in Spain using single-molecule array (SiMoA) and compared them with 53 healthy controls (HCs). We performed multivariable regression to analyse the association between sNfL levels and functional outcome at 1 year.

Results Patients with GBS had higher NfL levels than HC in serum (55.49 pg/mL vs 9.83 pg/mL, $p < 0.0001$) and CSF (1308.5 pg/mL vs 440.24 pg/mL, $p = 0.034$). Patients with preceding diarrhoea had higher sNfL than patients with respiratory symptoms or no preceding infection (134.90 pg/mL vs 47.86 pg/mL vs 38.02 pg/mL, $p = 0.016$). sNfL levels correlated with Guillain-Barré Syndrome Disability Score and Inflammatory Rasch-built Overall Disability Scale (I-RODS) at every timepoint. Patients with pure motor variant and Miller Fisher syndrome showed higher sNfL levels than patients with sensorimotor GBS (162.18 pg/mL vs 95.50 pg/mL vs 38.02 pg/mL, $p = 0.025$). Patients with acute motor axonal neuropathy had higher sNfL levels than other variants (190.55 pg/mL vs 46.79 pg/mL, $p = 0.013$). sNfL returned to normal levels at 1 year. High baseline sNfL levels were associated with inability to run (OR=1.65, 95% CI 1.14 to 2.40, $p = 0.009$) and lower I-RODS ($\beta -2.60$, 95% CI -4.66 to -0.54 , $p = 0.014$) at 1 year. Cut-off points predicting clinically relevant outcomes at 1 year with high specificity were calculated: inability to walk independently (>319 pg/mL), inability to run (>248 pg/mL) and ability to run (<34 pg/mL).

Conclusion Baseline sNfL levels are increased in patients with GBS, are associated with disease severity and axonal variants and have an independent prognostic value in patients with GBS.

INTRODUCTION

Guillain-Barré syndrome (GBS) diagnosis relies on clinical and electrophysiological criteria¹ and

albumino-cytological dissociation in cerebrospinal fluid (CSF), but patients differ considerably in their presentation, clinical course and prognosis. Previous prognostic models, based on clinical and epidemiological features, predict the ability to walk at 6 months^{2,3} or the probability to develop respiratory insufficiency.⁴

Neurofilament light chain (NfL) is becoming the most important axonal damage biomarker^{5,6} in neurology after the introduction of ultrasensitive techniques such as the single-molecule array (SiMoA).^{7,8} Older studies investigated neurofilament heavy chain (NfH) in GBS, showing increased CSF NfH levels.^{9,10} CSF NfL levels were also described to associate with GBS outcome in a small study.¹¹ A recent study showed increased serum and CSF NfL levels in acquired peripheral neuropathies, including five patients with GBS¹² and another study showed increased serum NfL (sNfL) levels in diverse diseases, including 19 patients with GBS, but clinical correlations were not performed.¹³ A very recent retrospective study showed that high sNfL levels associated with poor short-term prognosis, but the analysis did not take confounding variables in account.¹⁴ Thus, sNfL levels as a long-term prognosis biomarker in GBS have not been investigated so far.

Our study aims to: (1) describe NfL levels in serum and CSF from GBS and GBS-variant patients at baseline and at 1 year (serum only); and (2) analyse the relationship between baseline sNfL levels and prognosis at 1 year.

MATERIALS AND METHODS**Subjects, standards protocol approvals and patient consents**

We collected data, involving 98 patients enrolled in the Spanish cohort of the International Guillain-Barré Syndrome Outcome Study (IGOS) study.¹⁵ The IGOS is a prospective, observational cohort study including patients fulfilling diagnostic criteria of the National Institute of Neurological Disorders

and Stroke (NINDS)¹ or Miller Fisher syndrome (MFS) and other variants of GBS.^{16 17} There are no exclusion criteria.¹⁵ All patients were included within 2 weeks from disease onset. Patients from the Spanish cohort were enrolled between February 2013 and January 2019. Additionally, 53 sera from age-matched healthy controls (HCs) and 10 CSF samples from age-matched HCs were included. CSF samples from HC were obtained from Sant Pau Memory Unit Biobank. Serum and CSF samples were aliquoted and stored at -80°C until needed. All patients gave written informed consent to participate in the study.

SiMoA NfL measurements

Measurement of serum and CSF NfL levels was performed using the SIMOA Nf-light kit in SR-X immunoassay analyzer, Simoa (Quanterix Corp, Boston, Massachusetts, USA), which runs ultrasensitive paramagnetic bead-based enzyme-linked immunosorbent assays. Baseline samples were analysed in duplicates following the manufacturer's instructions and standard procedures. Samples were analysed at baseline and at 52 weeks (in patients with follow-up samples available; $n=33$). Sixty-one of the baseline samples analysed (62.2%) were collected before starting treatment. All NfL values were within the linear ranges of the assays. The intra-assay and inter-assay coefficients of variation at intermediate level (16.15 pg/mL) were 6.2% and 4.9%, respectively.

Data collection

Data were collected prospectively, including demographic features (age and gender) and disease variables: days since onset, infectious antecedent event, Medical Research Council (MRC) sumscore and disability scores, including the GBS Disability Score¹⁸ (GDS) and the Inflammatory Rasch-built Overall Disability Scale (I-RODS) scores¹⁹ at different timepoints. We also collected Brighton diagnostic criteria, ranging from levels 1 (highest level) to 4.^{16 20} Results of routine CSF examination, nerve conduction studies (NCS) and treatment were also collected. We defined an elevated CSF protein level as higher than 0.45 g/L.^{15 21} Clinical variants were defined as sensorimotor, pure motor, pure sensory, MFS, ataxic and pharyngeal-cervical-brachial variant.²² NCS results were classified as acute inflammatory demyelinating polyneuropathy (AIDP), acute motor axonal neuropathy (AMAN), acute motor-sensory axonal neuropathy (AMSAN), equivocal or normal. GDS was recorded at baseline and at 4, 26 and 52 weeks. I-RODS (highest score 48, indicating no disability)¹⁹ was assessed at 4, 26 and 52 weeks. The 'ability to run' variable was extracted from I-RODS, and the 'ability to walk independently' variable was extracted from GDS.

Statistical analysis

Descriptive statistics are shown as mean ($\pm$ SD) or median (IQR) in continuous variables and as frequencies (percentages) in categorical variables. sNfL levels were non-normally distributed as tested by the Shapiro-Wilk normality test. Thus, a logarithmic transformation of the variable was performed to approach the normal distribution (logNfL). We summarised NfL levels using geometric means (GeoMeans), GeoMean 95% CI and coefficient of variation. Comparisons between patients with GBS and HC were performed by the Wilcoxon rank-sum test. Kruskal-Wallis test was used to compare groups at baseline. Wilcoxon matched pairs signed rank test was used to compare baseline sNfL and sNfL at 52 weeks. We used Spearman's coefficient to assess correlation between variables.

To investigate the association between sNfL and prognosis, we performed two types of multivariable regression analyses. First, we conducted a multivariable logistic regression analysis to predict the ability to run at 1 year of follow-up. Second, we performed a multivariable linear regression analysis to investigate the association between logNfL and I-RODS scale at 1 year. In both analyses, we performed a stepwise backward regression modelling to select variables independently associated with the outcome. The statistical analyses performed and the variables introduced in our multivariable models were predefined based on known prognostic factors (age, GDS, diarrhoea, MRC sumscore at 1 week and AMAN^{23 24}). We report two different models to avoid collinearity between the variables AMAN and MRC sumscore at 1 week. To perform the multivariable analysis, we excluded patients with MFS, because our aim was to predict GBS prognosis and MFS is considered a different disease, including different pathophysiology, clinical presentation (it does not present with weakness), treatment (often untreated) and outcome (considered self-limiting and benign). The final models were adjusted by potential confounders. A significant confounding effect was defined as an absolute change $>10\%$ in the regression coefficients when introducing the variable into the model.

ORs for the logistic regression analysis and beta coefficients (β) for the linear regression analysis were reported with 95% CIs and p values.

Additionally, we evaluated the predictive capacity of NfL levels in the acute phase and at 1 year (residual disability) for five different clinically relevant endpoints, established prior to the analysis: (1) ability to run at 1 year, (2) inability to run at 1 year, (3) inability to walk without assistance at 1 year, (4) ventilation and (5) death. First, we performed a univariate logistic regression analysis for each endpoint using NfL levels as the exposure variable. Then, we performed a receiver operating characteristic (ROC) analysis comparing the predictions against the endpoint and, eventually, we selected the sNfL cut-off points that better predicted the endpoint (aiming for highest specificity). Finally, we tested the predictive capacity of each cut-off point using multivariable logistic regression analyses adjusting for age and AMAN as possible confounders. OR with 95% CI, area under the curve (AUC), sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) for each cut-off point were reported.

Statistical significance for all analyses was set at 0.05 (two-sided). The analysis was carried out in GraphPad Prism V.8 and Stata V.15.

Data availability

Anonymised data not published within this article will be made available by request from any qualified investigator.

RESULTS

Baseline characteristics

We enrolled 98 participants from 11 Spanish centres. Patients with GBS had an average of 57.4 years and were predominantly men (57%); 68.4% of patients presented an antecedent infectious event. The median time from symptom onset to inclusion was 4 days (IQR 3–6). Most patients had an initial GDS between 2 and 4 (93.9%). CSF was examined in 90 (91.8%) patients within a median time of 3 days (IQR 2–6) since disease onset. High CSF protein levels were detected in 68.9% of the patients. Most patients were classified in level 1 of diagnostic certainty for GBS following Brighton criteria (65.9%), 33% of patients were

Table 1 Baseline serum and CSF NfL in patients with GBS and HCs

	GBS patients	HC	P value
Age			
Mean±SD	57.4±16.6	53.1±14.8	0.072
Gender			
Male, n (%)	56 (57.1)	23 (43.4)	0.51
Serum NfL (pg/mL)			
n	98	53	<0.001
GeoMean	55.49	9.83	
Median	40.00	7.93	
Max	3573.03	56.82	
Min	4.22	4.30	
CSF NfL (pg/mL)			
n	24	10	0.034
GeoMean	1308.50	440.24	
Median	883.57	493.79	
Max	32915.47	1469.97	
Min	212.74	139.42	

CSF, cerebrospinal fluid; GBS, Guillain-Barré Syndrome; GeoMean, geometric mean; HCs, healthy controls; sNfL, Serum neurofilament light chain.

classified in level 2 and one patient in level 3. Sixty-six per cent of patients presented with the typical sensorimotor variant and most patients were classified as AIDP (59.2%) and were treated with intravenous immunoglobulin (IVIg) (77.6%) or IVIg plus plasma exchange (10.2%).

Association of baseline sNfL with disease characteristics

Patients with GBS had significantly higher sNfL levels (55.49 pg/mL) than healthy controls (HCs) (9.83 pg/mL, $p<0.0001$, table 1, figure 1A). This difference was also observed in CSF (1308.5 pg/mL vs 440.24 pg/mL, $p=0.034$, figure 1B). We found a correlation between sNfL and CSF NfL ($r=0.62$, $p<0.001$, online supplemental figure 1A).

sNfL levels correlated with age in HC ($r=0.64$, $p<0.001$) but not in patients with GBS ($r=0.18$, $p=0.07$) (online supplemental figure 1B). Patients with diarrhoea had higher sNfL levels than those patients without any infectious antecedent or with respiratory symptoms (134.9 pg/mL vs 38.02 pg/mL vs 47.86 pg/mL; $p=0.016$). CSF protein levels were not correlated with sNfL or CSF NfL. sNfL and CSF NfL levels did not correlate with time since symptom onset.

Patients with pure motor GBS variant or MFS showed higher sNfL levels than patients with typical GBS (162.18 pg/mL vs 95.5 pg/mL vs 38.02 pg/mL, respectively; $p=0.025$; figure 2A). When we stratified patients according to NCS, patients with AMAN had higher sNfL levels than other patients (199.53 pg/mL vs 46.77 pg/mL, $p=0.006$) and patients with equivocal NCS and AMSAN had higher sNfL levels than AIDP (figure 2B). The seven patients with normal NCS showed higher sNfL levels than AIDP patients (53.7 pg/mL vs 35.48 pg/mL, table 2). Of those patients with normal NCS, five were patients with MFS, one had a pure sensory variant and one patient was a typical GBS (89.13 pg/mL vs 10.47 pg/mL vs 23.44 pg/mL, respectively). Patients that did not receive treatment due to mild symptoms had normal levels of sNfL. Further information about sNfL levels and disease characteristics is detailed in table 2.

Association of baseline sNfL with clinical scales: sNfL levels clearly correlated with the GDS and I-RODS scores. sNfL levels tend to be higher with every GDS increase ($r=0.313$, $p=0.002$; table 2, figure 3). We found a correlation with GDS at every time-point and with maximum GDS achieved ($r=0.251$, $p=0.013$) (online supplemental table 1). Baseline sNfL levels displayed a negative correlation with the I-RODS scale at 4, 26 and 52 weeks (online supplemental table 1).

sNfL levels at 1 year

We analysed sNfL levels in 33 patients at baseline and at week 52. sNfL levels returned to normal control levels at 52 weeks in all patients that had high baseline NfL levels; sNfL levels did not change in patients with normal baseline levels (online supplemental figure 2).

Association of baseline sNfL and prognosis

Approximately, 71% and 74.5% of patients could walk independently at 6 months and at 1 year, respectively, and 60% and 67% of patients could run at 6 months and at 1 year, respectively. Ten patients needed mechanical ventilation during a median time of 11 days (IQR 8–33); four patients died. Two patients died due to respiratory insufficiency, one patient due to pneumonia and one patient because of cancer progression. Baseline sNfL levels were associated with the ability to run at 1 year (GeoMean sNfL levels 33.11 pg/mL vs 123.03 pg/mL; $p<0.001$). Baseline sNfL levels were higher in patients that died (GeoMean sNfL levels 53.70 pg/mL vs 229.09 pg/mL, $p=0.11$) or needed mechanical

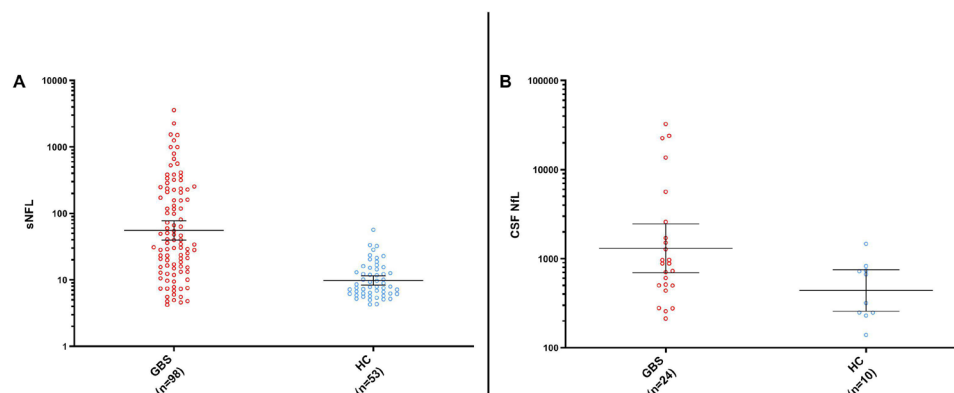


Figure 1 Baseline NfL in patients with GBS versus HCs in serum (A) and in CSF (B). The line in the centre represents the GeoMean value and the whiskers indicate the 95% CI. CSF, cerebrospinal fluid; GBS, Guillain-Barré Syndrome; GeoMean, geometric mean; HCs, healthy controls; sNfL, serum neurofilament light chain.

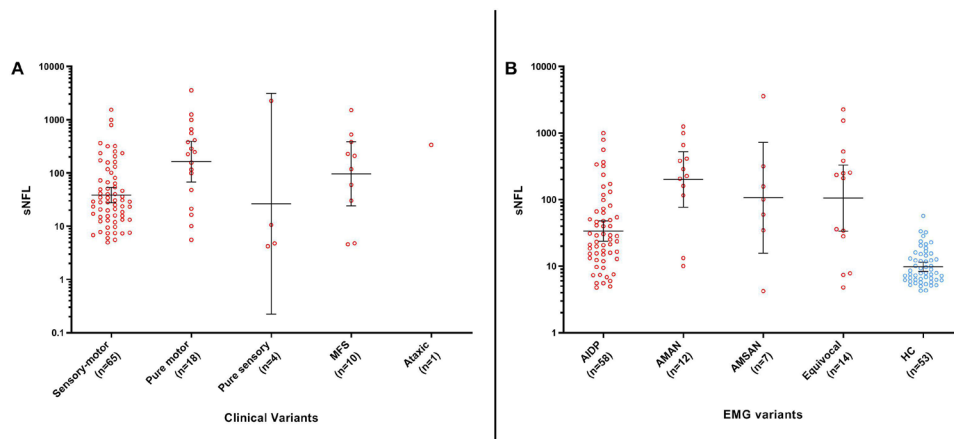


Figure 2 (A) sNfL levels in clinical variants. (B) sNfL and EMG classification. The line in the centre represents the GeoMean value and the whiskers indicate the 95% CI. AIDP, acute inflammatory demyelinating polyneuropathy; AMAN, acute motor axonal neuropathy; AMSAN, acute motor-sensory axonal neuropathy; EMG, electromyography; GeoMean, geometric mean; HC, healthy control; MFS, Miller Fisher Syndrome; sNfL, serum neurofilament light chain.

ventilation (GeoMean sNfL levels 52.48 pg/mL vs 83.17 pg/mL, $p=0.446$), but these differences were not statistically significant.

In the univariate analysis, logNfL, AMAN, age and initial GDS were associated with the ability to run at 1 year and logNfL, age and initial GDS were associated with the I-RODS at 1 year (table 3). In the multivariable logistic regression analysis, which included four potential confounder variables in the initial model (age, initial GDS, AMAN and diarrhoea), higher baseline NfL levels were independently associated with the inability to run at 1 year after a backward stepwise selection modelling (OR=1.65, 95% CI 1.14 to 2.40, $p=0.009$; table 3). Undoing the log transformation of the variable, this increase represents an OR=1.019; 95% CI 1.037 to 1.002, for each 10 pg/mL of sNfL. In the multivariable linear regression analysis to predict I-RODS at 1 year ($\beta=-2.60$, 95% CI -4.66 to -0.54 ; $p=0.014$) (see table 3).

We additionally repeated the two multivariable analyses (logistic and linear) with and without patients with MFS. We mainly focused on patients with GBS because MFS is considered a different disease and the aim of the study was to assess baseline sNfL to predict GBS outcome. Nonetheless, the inclusion of MFS in the multivariable analysis models did not change results significantly (online supplemental table 2).

We also performed multivariable logistic regression analyses including MRC sumscore at 1 week as confounder and found that higher baseline NfL levels were independently associated with the inability to run (OR=1.89, 95% CI 1.20 to 2.80, $p=0.001$) and with lower I-RODS scores at 1 year ($\beta=-2.30$, 95% CI -3.95 to -0.67 ; $p=0.006$) (online supplemental table 3).

Ability of sNfL levels to predict clinically relevant outcomes

We finally evaluated the ability of baseline sNfL levels to predict the outcome in five clinically relevant endpoints: (1) ability to run at 1 year, (2) inability to run at 1 year, (3) inability to walk without assistance at 1 year, (4) ventilation and (5) death. We performed an ROC analysis for each endpoint selecting the cut-off points with the highest specificity (see Methods section). sNfL levels higher than 319 pg/mL discriminated patients unable to walk independently at 1 year (OR=5.20, 95% CI 1.02 to 26.34, $p=0.047$, specificity 89.4%, sensitivity 33.3%); sNfL levels higher than 248 pg/mL discriminated patients unable to run at

1 year (OR=6.81, 95% CI 1.64 to 28.21, $p=0.008$, specificity 94.2%, sensitivity 39.3%) and sNfL levels lower than 34 pg/mL predicted complete recovery, defined as the ability to run at 1 year (OR=6.59, 95% CI 2.02 to 21.46, $p=0.002$, specificity 82.1%, sensitivity 69.2%). We were unable to define good prognostic sNfL level cut-off points for ventilation and death due to the low number of patients. AUC, PPV and NPV for each cut-off point are detailed in online supplemental table 4. The three cut-off points maintain their significance evaluating I-RODS at 1 year by linear regression (online supplemental table 4).

DISCUSSION

Our study shows that sNfL levels correlate with diverse clinical, epidemiological and electrophysiological features and associate with clinical outcomes independently of other known prognostic variables. Moreover, sNfL levels can be used to set different cut-off points that discriminate patients at risk of developing mild (inability to run) or moderate (ability to walk independently) disability as well as those recovering completely (ability to run) at 1 year. As expected, due to the monophasic nature of GBS and the ability of sNfL to only detect ongoing axonal damage, all patients with high sNfL levels at baseline returned to normal levels at 1 year.

Baseline characteristics in patients with GBS and prognostic values (ability to walk at 6 months and 1 year) in our cohort are similar to other studies, and similar to the IGOS global dataset,²⁵ except for an increased proportion of patients with AMAN (12.2%) and MFS (10.2%). This could be a selection bias due to the higher severity of patients with AMAN or the rarity of MFS variant.

We found a correlation between baseline sNfL levels and disease severity, in agreement with previously reported findings in other peripheral neuropathies, such as chronic inflammatory demyelinating polyneuropathy,²⁶ Charcot-Marie-Tooth disease²⁷ or neuropathy in hereditary transthyretin amyloidosis.²⁸ The median levels of sNfL in patients with GBS were higher than the sNfL levels reported in other peripheral neuropathies, but there is a high variability among patients with GBS, ranging from normal levels to levels 100-fold higher than HC. This variability is partially explained by the different clinical and electrophysiological variants: sNfL levels are higher in axonal variants (AMAN). Other factors, not considered in this study, could have influenced heterogeneity in sNfL levels (treatment used,

Table 2 Relationship between sNfL levels and basal characteristics

Basal characteristics (n=98)		sNfL (pg/mL); GeoMean (GeoMean 95% CI), CV%	P value
Age, mean±SD	57.35±16.6	–	0.071
Gender, n (%)			
Male	56 (57.1)	52.48 (31.62 to 79.43), 41.5	0.625
Female	42 (42.9)	60.26 (39.81 to 100), 41.8	
Infectious antecedent, n (%)			
Diarrhoea	24 (24.5)	134.90 (63.10 to 251.19), 36.6	0.016
Respiratory infection	43 (43.9)	38.02 (25.12 to 63.10), 42.6	
None	31 (31.6)	47.86 (28.35 to 81.50), 39.3	
Evolution time (days), median (IQR)	4 (3–6)	–	0.413
Initial GDS, median (IQR)			
1, n (%)	4 (4.1)	6.17 (3.98 to 10), 25.4	0.005
2, n (%)	28 (28.6)	35.48 (19.95 to 63.10), 46.1	
3, n (%)	19 (19.4)	64.56 (31.62 to 125.89), 36.5	
4, n (%)	45 (25.9)	77.62 (59.12 to 125.89), 35.7	
5, n (%)	2 (2)	375.83; 53.5	
MRC sumscore, median (IQR)			
Baseline	50 (42–56)	–	<0.001
At 1 week	51 (39–58)	–	0.073
GBS variants, n (%)			
Sensorimotor	65 (66.3)	38.02 (25.12 to 50.12); 37	0.009
Pure motor	18 (18.4)	162.18 (79.43 to 398.11); 34.7	
Pure sensory	4 (4.1)	26.30 (1.26 to 501.19); 91.6	
MFS	10 (10.2)	95.50 (31.62 to 316.23); 42.3	
Ataxic	1 (1)	338.84	
EMG classification, n (%)			
AIDP	58 (59.2)	33.88 (25.12 to 50.12), 37.7	0.008
AMAN	12 (12.2)	199.53 (79.43 to 501.19), 28.5	
AMSAN	7 (7.1)	107.15 (25.12 to 501.19), 44	
Equivocal	14 (14.3)	104.7 (39.81 to 316.22), 43	
Normal	7 (7.1)	53.70 (12.59 to 251.19), 50	
Treatment, n (%)			
No	6 (6.1)	7.59 (3.98 to 12.59), 37.5	0.009
IVIg	76 (77.6)	72.44 (50.12 to 100.43), 39.1	
IVIg x2	5 (5.1)	32.36 (15.85 to 79.43), 27	
IVIg+PLEX	10 (10.2)	30.90 (12.59 to 79.43), 41.8	
PLEX	1 (1)	117.49	

AIDP, acute inflammatory demyelinating polyneuropathy; AMAN, acute motor axonal neuropathy; AMSAN, acute motor-sensory axonal neuropathy; CV, coefficient of variation; EMG, Electromyography; GBS, Guillain-Barré Syndrome; GDS, Guillain-Barré Syndrome Disability Score; GeoMean, geometric mean; IVIg, intravenous immunoglobulin; MFS, Miller Fisher Syndrome; MRC, Medical Research Council; PLEX, plasma exchange; sNfL, serum neurofilament light chain.

comorbidities, potential misdiagnosis with GBS mimics, physical therapy regimens) and will need to be assessed in larger cohorts. Some of the samples were collected after starting treatment with IVIg or PLEX; thus, the possible effect of treatment in sNfL should be assessed in future studies.

We also found elevated sNfL levels in patients with MFS despite their benign prognosis (87.5% were able to run at 6 months and their median I-RODS was 48). When we analysed the NCS for patients with MFS, we found five patients with normal NCS, four patients with equivocal NCS and one patient with AMSAN, classified as MFS-overlap syndrome. The underlying mechanism involved in the sNfL elevation in patients with MFS despite unremarkable peripheral nerve electrophysiological alterations is unclear: one possible explanation could be that either preganglionic nerve roots (difficult to assess with routine electrophysiology) or cerebellar damage are driving this sNfL increase instead of peripheral nerve damage,^{29–31} but larger cohorts with MFS (including CSF) are needed as a non-apparent bias may be driving this unexpected results.

Smaller studies have previously studied neurofilament levels in serum and CSF^{9–13} and some of them found an association of poor outcomes with severity.¹⁴ However, we provide the first large prospective study assessing the relationship of sNfL (tested with SiMoA) with GBS features and long-term prognosis. We found a clear relationship between sNfL and prognosis: high baseline sNfL was associated with inability to run and with I-RODS scores at 1 year in a multiple logistic and linear regression models, adjusted by known prognostic factors. We also aimed to establish cut-off points to predict clinically relevant outcomes with high specificity. These three cut-off points allowed us to predict that in our sample: (1) a patient with baseline sNfL >319 pg/mL is not going to be able to walk independently at 1 year with almost 90% specificity; (2) a patient with baseline sNfL >248 pg/mL is not going to be able to run at 1 year with a 94.2% of specificity and (3) a patient with baseline sNfL <34 pg/mL is going to be able to run at 1 year with a 82.1% of specificity. Considering that most patients with GBS are going to walk independently at 1 year anyway, early recognition of

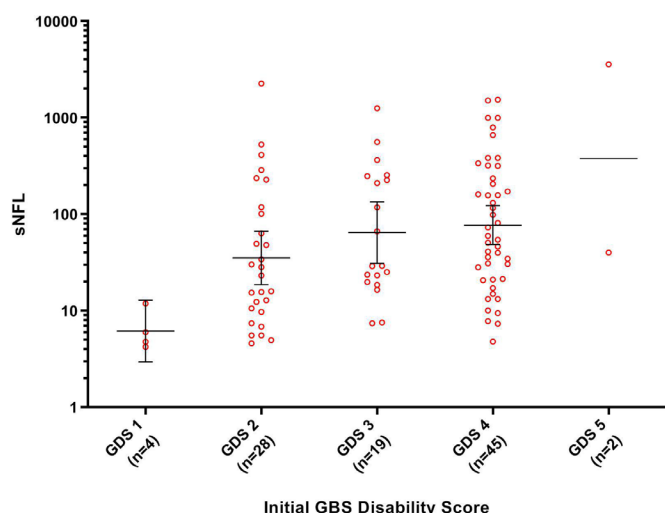


Figure 3 Correlation between initial GDS and baseline sNfL levels. The line in the centre represents the geometric mean value and the whiskers indicate the 95% CI. GBS, Guillain-Barré Syndrome; GDS, Guillain-Barré Syndrome Disability Score; sNfL, serum neurofilament light chain.

Table 3 Association between baseline NfL levels and prognosis

Univariate analysis for ability to run at 1 year					
	OR	95% CI	P value		
Variable					
logNfL	1.90	1.34 to 2.69	<0.001		
AMAN	6.53	1.57 to 27.17	0.010		
Diarrhoea	2.33	0.83 to 6.57	0.109		
Age	1.04	1.01 to 1.07	0.023		
Initial GDS	1.81	1.08 to 3.04	0.025		
Univariate analysis for I-RODS at 1 year		Beta coefficient	95% CI	Std beta	P value
Variable					
logNfL	−3.67	−5.65 to −1.69	−0.40	<0.001	
AMAN	−6.14	−16.48 to 4.19	−0.14	0.240	
Diarrhoea	−6.03	−14.41 to 2.34	−0.17	0.155	
Age	−0.27	−0.47 to −0.08	−0.31	0.007	
Initial GDS	−5.54	−8.63 to −2.44	−0.39	0.001	
Multivariable logistic analysis for ability to run at 1 year*					
	OR	95% CI	P value		
Variable					
logNfL	1.65	1.14 to 2.40	0.009		
AMAN	6.19	1.01 to 38.02	0.049		
Age	1.05	1.01 to 1.09	0.022		
Multivariable linear analysis for I-RODS at 1 year†					
	Beta	95% CI	Std beta	P value	
Variable					
logNfL	−2.60	−4.66 to −0.54	−0.29	0.014	

*Results from the multivariable logistic regression analysis, and †results from the multivariable linear regression analysis after a backward stepwise selection modelling. The variables included in the initial model were age, initial GDS, AMAN and diarrhoea.

AMAN, acute motor axonal neuropathy; GDS, Guillain-Barré Syndrome Disability Score; I-RODS, Rasch-built Overall Disability Scale; logNfL, log-transformed neurofilament light chain; Std, standardised.

patients with poor and excellent prognoses could eventually help physicians guide treatment choices in the future and provide better prognostic information for patients in the recovery phase.

We have performed all analyses using baseline sNfL levels. However, sNfL levels may increase during hospital admission and it is possible that sNfL levels measured at 1 or 2 weeks were higher and able to predict the long-term outcome better. However, we performed sNfL levels at baseline because our aim was to assess the predictive ability of sNfL when the patient is admitted, to make prognostic inferences and inform care choices the earliest possible.

Nowadays sNfL testing is not readily available for clinical practice (although it is increasingly available for routine testing in neurodegenerative diseases), but incorporating sNfL levels in the existing prognostic models for GBS²⁻⁴ could be an option to improve the GBS prognosis prediction in the future. Our study provides one of the largest prognostic studies in GBS, but larger studies should confirm our findings and establish optimised sNfL cut-off points for relevant clinical endpoints in more diverse populations.

Unlike other diseases, such as multiple sclerosis,³² differences in NfL levels between patients with GBS and healthy controls are relatively lower in CSF than in serum. These results are surprising since nerve root involvement (detected by MRI imaging or neurophysiological studies) is prominent in GBS. The immune response in GBS also targets the peripheral nerves, and the amount of affected axons may be much higher in the periphery. In addition, some authors propose that preforaminal root inflammation may result in lower axonal damage than in postforaminal roots because the absence of epi-perineurium allows higher tissue compliance in intrathecal nerve segments.^{33,34} However, due to the low number of patients in whom we have measured the NfL in CSF, these are hypotheses that need to be interpreted with caution.

An important issue in GBS and other inflammatory neuropathies is the lack of biomarkers that could be used as surrogate markers for disease activity and as secondary endpoints in clinical trials. Clinical trials in inflammatory neuropathies, including GBS, typically use clinical scales as primary and secondary endpoints. Moreover, the traditional primary endpoint (ability to walk unaided at 6 months) may not be sensitive enough to detect treatment effect in GBS trials.³⁵ The relationship of sNfL levels with disease severity, axonal damage and prognosis suggests that sNfL levels could be an informative predictor of treatment effect in routine practice or in clinical trials, as it happens in other diseases.³⁶

One of the limitations of our study is the lack of exclusion criteria in the IGOS study: all patients with GBS are included in IGOS and data on previous neurologic diseases influencing NfL levels or clinical outcomes were not recorded. In our sample, we have seven patients that have comorbidities affecting mobility, but we did not exclude those patients from multivariable analyses to strictly follow IGOS study inclusion and exclusion criteria. Another point to study would be the time to recovery of patients with high sNfL versus patients with low sNfL. To evaluate this, it would be necessary to perform more frequent or self-registered visits informing about relevant outcomes, since there were no visits scheduled between the 24-week and 52-week timepoints. Furthermore, we are aware that our established cut-off points are valid only for our sample of patients. Future, larger studies, with repeated sampling, and addressing short-term dynamics of the sNfL levels could study the relationship of sNfL levels with treatment response, best time points to test sNfL for prognostic purposes and better representative cut-off points for clinically

relevant endpoints. However, our study provides the proof of concept to support the use of the sNfL cut-off points as prognostic predictors in GBS in the future.

In summary, our study proves that sNfL are increased in patients with GBS, that they correlate with disease severity and axonal damage and that they could be used as an informative prognostic biomarker in patients with GBS.

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Contributors LM-A contributed to acquisition of data, performed the experiments, analysed the data and drafted the manuscript for intellectual content. PC-R had a major role in analysing the data and performing all statistical analyses. MCL, EPG, JD-M, RR-G, EC-V, CC, CH, GG-G, MCJ-M, JB, MJS, TG-S, JP, CM, IR-M, JJ-P, EM-H, GM and CDG collected the samples and data and revised the manuscript for intellectual content. LM and DA helped to perform the experiments and revised the manuscript for intellectual content. ND, EG, AL and II revised the manuscript for intellectual content. LAQ designed and conceptualised the study, interpreted the data and revised the manuscript for intellectual content.

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Competing interests LAQ has provided expert testimony for Grifols, Sanofi-Genzyme, Novartis, UCB, Roche and CSL Behring and received research funds from Novartis Spain, Sanofi-Genzyme and Grifols. LM-A has received speaking honoraria from Roche. EP-G has received speaking honoraria from Roche and Biogen. JD-M has provided expert testimony for PTC and Sanofi-Genzyme, has been external advisor for Sanofi, Sarepta and Audentes and received research funds from Sanofi-Genzyme and Boehringer. DA participated in advisory boards from Fujirebio-Europe and Roche Diagnostics and received speaker honoraria from Fujirebio-Europe, Nutricia and from Krka Farmacéutica S.L. GG-G has received speaking honoraria from Sanofi-Genzyme, Takeda and has provided expert testimony for Biogen and CSL Behring. The other authors report no disclosures.

Patient consent for publication Not required.

Ethics approval The study general IGOS protocol and the study of biomarkers (including NfL) were approved by the Ethics Committee of the Hospital de la Santa Creu i Sant Pau (IGOS protocol, code 12/142; biomarker/NfL research in GBS, codes

17/033 and 20/034) and the local Institutional Review Boards of participating hospitals or universities.

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